

2022 S.T. Yau High School Science Award (Asia)

Research Report

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Title of Research Report

An Investigation on the aerodynamics of flight of a hand helicopter

Date

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Abstract

A simple hand helicopter can be made by attaching rotor blades to one end of a vertical stick. The helicopter moves upwards when the stick is spun above a threshold angular velocity and released. Such toys are commonly found in toy shops. In this study, computational fluid dynamics is used to predict the air flow around a hand helicopter from which the lift and drag acting on the hand helicopter's blades can be obtained. Newton's 2nd Law is then applied to find the hand helicopter's equations of motion. These are solved to predict the helicopter's trajectory and maximum height. The angle of attack, length of blades and thickness of blades are varied and their effects on the lift coefficient, drag coefficient, lift-off and maximum height are investigated. These results offer insight into the parameters which will allow a hand helicopter achieve its optimal maximum height. The conditions required for the hand helicopter to fly stably are also investigated and explained using the Tennis Racket Theorem.

Keywords: Aerodynamics, Tennis Racket Theorem, Hand Helicopter, Computational Fluid Dynamics

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Commitments on Academic Integrity and Honesty

Research Report

2022 S.T. Yau High School Science Award (Asia)

Commitments on Academic Honesty and Integrity

We hereby declare that we

1. are fully committed to the principle of honesty, integrity and fair play throughout the competition.
2. actually perform the research work ourselves and thus truly understand the content of the work.
3. observe the common standard of academic integrity adopted by most journals and degree theses.
4. have declared all the assistance and contribution we have received from any personnel, agency, institution, etc. for the research work.
5. undertake to avoid getting in touch with assessment panel members in a way that may lead to direct or indirect conflict of interest.
6. undertake to avoid any interaction with assessment panel members that would undermine the neutrality of the panel member and fairness of the assessment process.
7. observe the safety regulations of the laboratory(ies) where we conduct the experiment(s), if applicable.
8. observe all rules and regulations of the competition.
9. agree that the decision of YHSA(Asia) is final in all matters related to the competition.

We understand and agree that failure to honour the above commitments may lead to disqualification from the competition and/or removal of reward, if applicable; that any unethical deeds, if found, will be disclosed to the school principal of team member(s) and relevant parties if deemed necessary; and that the decision of YHSA(Asia) is final and no appeal will be accepted.

(Signatures of full team below)☒ 

Name of team member: Lim Li Xin Ted

☒

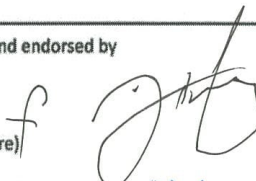
Name of team member:

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Name of team member:

☒Name of supervising teacher:  Tan Guan Seng

Noted and endorsed by

(signature) 

Name of school principal: Soh Lai Leng Magdalen

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1 Introduction

Aerodynamics is an important field of physics governing the flow of air around objects, resulting in the forces of lift and drag. A good understanding of aerodynamics is critical to designing efficient planes and helicopters that can move faster and further with lower fuel cost. With the advent of drones and other small flying vehicles, the aerodynamics of small flying objects such as the hand helicopter has become ever more important. The methods used in this paper can be extended to model the motion of other small flying objects such as drones, to provide more precise control over their flight.

This paper reports on the use of 2 dimensional and 3 dimensional simulations to predict the lift and drag of a hand helicopter in Section 2 and the comparison of these predictions with experimental data. Newton's 2nd Law and the lift / drag equation is used in Section 3 to obtain the equations of motion of the hand helicopter and an analytical expression for the minimum angular velocity for lift-off. The theoretical predictions for maximum height and minimum angular velocity for lift-off were then compared with experimental data. The angle of attack, thickness and length of the helicopter blades were varied. Finally, the stability of the hand helicopter (which has not been previously investigated comprehensively in prior literature) is analysed with the Tennis Racket Theorem in Section 4. In Section 5, based on the results obtained from our analyses, we discuss the various parameters which can be optimised to allow the hand helicopter to achieve its maximum height.

1.1 Literature Review

Prior studies by Nakamura et al. [1] analysing the lift and drag produced by the hand helicopter have made use of machine learning algorithms based on experimental data to predict the lift and drag coefficients for blades of different shapes. Kawamura et al. [2] used computational fluid dynamics to find the lift and drag coefficients of the hand helicopter. However, Kawamura et al. [2] made use of simulated low Reynolds number flows which are not applicable to our hand helicopters. This paper will investigate high Reynolds number flows and thus, requires the modelling of turbulence with computational fluid dynamics.

To analyse the motion of the hand helicopter, Nakamura et al. [1] integrates across the blade of the hand helicopter to obtain the lift and drag force. Following this, Newton's 2nd Law is used to obtain the equations of motion of the hand helicopter and solve for its motion. We use similar methods which have been adapted to model the motion of our hand helicopter.

The launching of the hand helicopter is also an important part of this investigation. Various methods have been proposed to launch a hand helicopter. However, some designs such as that built by Takato et al. [3] require precision tools and machinery which are not readily available in a high school laboratory. In their study, Nakamura et al. [1] launched the hand helicopter through non-motorised means. However, this may not launch the hand helicopter straight upwards consistently. A holder specially designed by Taylor [4] to hold the hand helicopter and rotated with a motor was found to be effective to launch the hand helicopter straight upwards and with a consistent initial angular velocity. We have adapted the design of the holder to launch the hand helicopter in our experiments.

Takato et al. [5] teaches students to use computer-aided design and computational fluid dynamics through the design and construction of a bamboo taketombo. In our project, we will 3D print hand helicopters for our experiments as this allows us to easily customize the parameters of the hand helicopter for investigation.

2 Computational Fluid Dynamics

Lift and drag forces on the hand helicopter can be modelled via the lift and drag equations where F_L is the lift force and F_D is the drag force.

$$F_L = \frac{1}{2} C_L \rho A v^2 \quad (1)$$

$$F_D = \frac{1}{2} C_D \rho A v^2 \quad (2)$$

We seek to obtain the lift coefficient, C_L and the drag coefficient, C_D using computational fluid dynamics simulations because it is not possible to find these analytically due to the high Reynolds number of the system, leading to turbulent flow.

2.1 2D Modelling

To find the lift and drag acting on a wing, we first attempt to use a 2 dimensional cross-section. The Navier-Stokes equations are solved with the COMSOL Multiphysics Computational Fluid Dynamics (CFD) Module to find the air flow around the wing.

In order to model the turbulence around the wing, we use the Algebraic y-Plus Turbulence model because it is less computationally intensive and hence, the simulation is faster. Other turbulence models were tested such as the k-epsilon ($k - \epsilon$) model but there was no significant difference in the results when using different turbulence models.

From there, we can integrate the force exerted by the air on the wing and split it into lift and drag, from which the lift and drag coefficients can be obtained. The predictions are shown in Figure 1.

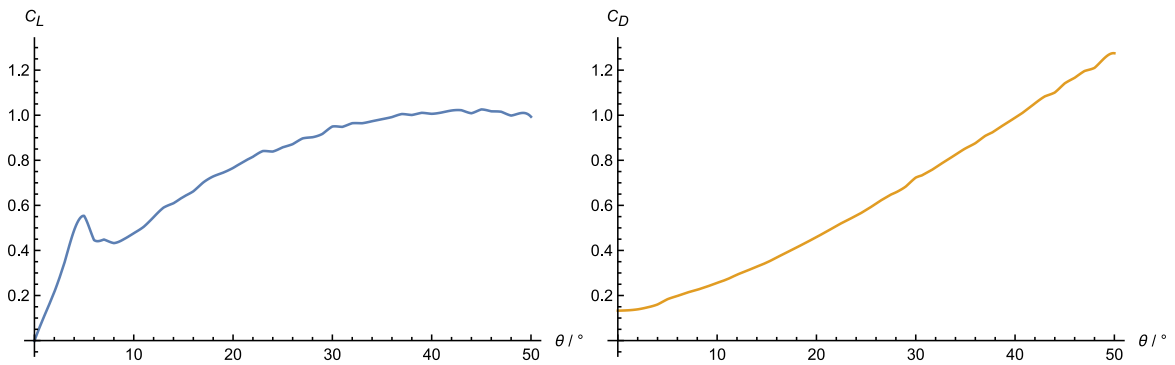


Figure 1: (left) C_L predicted by 2 dimensional model against angle of attack ϕ , (right) C_D predicted by 2 dimensional model against angle of attack ϕ

However, such a trend with the lift increasing rapidly, and then decreasing is not consistent with what is seen in our experiments (see Figure 6). This suggests that the 2 dimensional model is insufficient to accurately predict the lift and drag of the hand helicopter due to the presence of aerodynamic interference when the wings of the hand helicopter leave a turbulent wake behind them (see Figure 7). The turbulent air flow will affect the lift and drag acting on the wing, thus giving rise to the trend observed in the experiments.

2.2 3D Modelling

Since a 2 dimensional model is not able to account for aerodynamic interference, we have developed a full 3 dimensional model. Using the COMSOL Multiphysics CFD Module, we once again solve the Navier-Stokes equations but this time, in 3 dimensions.

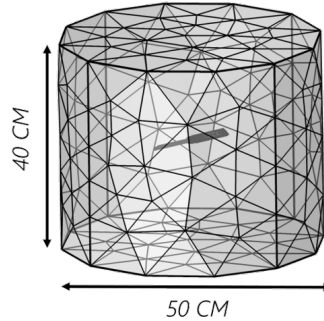


Figure 2: The geometry of the system in the 3D simulation. More information on the mesh used can be found in Appendix.

For the simulation, the helicopter blades are placed in a cylindrical column of air. The geometry of the system can be seen in Figure 2. The boundaries of the column have a boundary condition of no normal stress to simulate an infinite expanse of air. This mimicks the conditions that the blades will be in when the hand helicopter is flying.

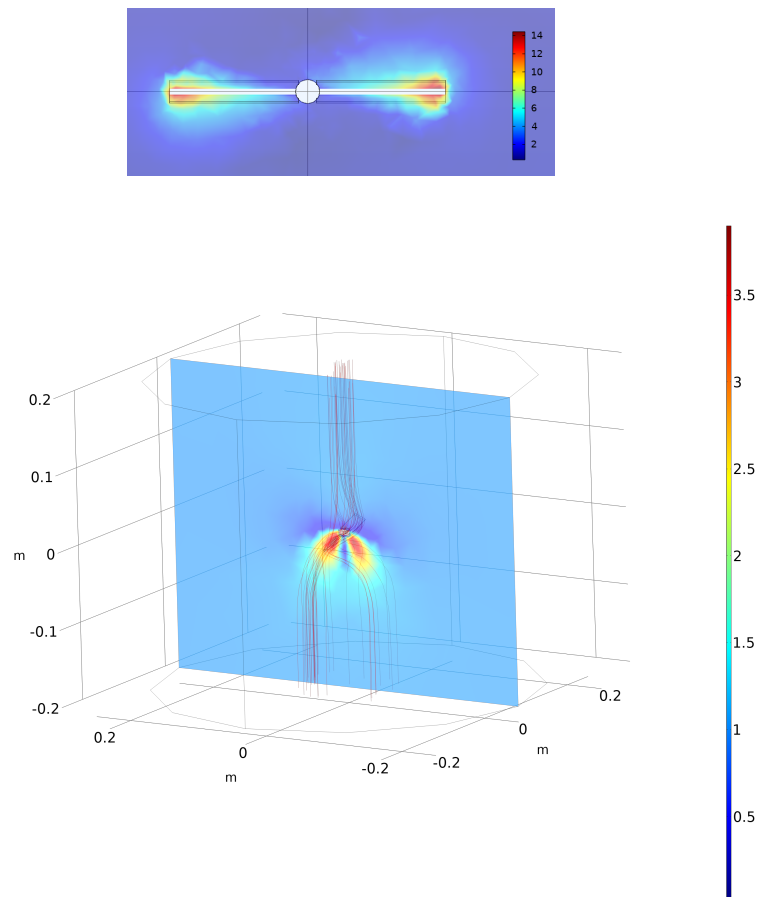


Figure 3: The velocity profile of the air around the hand helicopter blades. They are rotating anti-clockwise.

There is also a no-slip condition at the surface of the helicopter blades. The rotation of the helicopter blades is simulated using both fictitious rotational body forces and by using a moving mesh. There is no difference in the simulation results in both cases. The roughness of the helicopter blades is also accounted

for by adjusting the wall function to account for friction in the way suggested by [6]. However, there is little difference between the results for rougher and smoother blades in the simulation, thus showing that skin friction drag is insignificant.

The air velocity profile obtained from the simulation can be seen in Figure 3. The helicopter blades are rotating anti-clockwise. The source of the lift can be explained by looking at the streamlines, which show that the air is deflected downwards by the blades. This deflection of air is caused by the rotation of the hand helicopters. As a result of the air being deflected downwards, upward force acts on the blades due to Newton's 3rd Law which is known as the lift.

2.3 Experimental Comparison

For our experiments, 3D printed hand helicopters are used as they allow for easy modification of the helicopter's parameters such as their length, thickness and angle of attack. The 3D printed helicopter blades are rectangular with a constant width along their length for easier modelling (Figure 4).



Figure 4: A diagram of the 3D model of the hand helicopter blades.
(left) top view (right) side view

To find the lift coefficient experimentally, the blades are attached to a motor by a 3D-printed adapter and placed on a weighing scale to measure the lift produced (Figure 5 (left)). Cardboard is placed between the helicopter blades and the weighing scale to prevent the air flow around the blades from affecting the weighing scale reading. Wooden blocks are also used to elevate the motor to reduce the influence of ground effect on the experimental results.

To measure the drag coefficient, the propeller is attached to a ball bearing clamped by a retort stand and given an initial spin (Figure 5 (right)). The motion is then recorded with a high speed camera. A red / black dot is placed on the helicopter blades to help track its motion. The motion tracking is done with OpenCV.

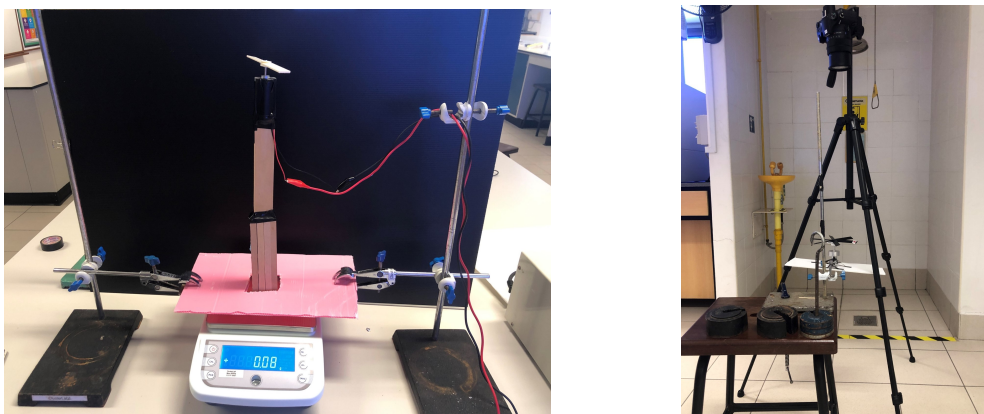


Figure 5: (left) The setup to measure the lift coefficient. (right) The setup to measure the drag coefficient.

We then compare the theoretical predictions with the data obtained from experiments when the angle of attack is varied. The good agreement between the computational fluid dynamics simulation and the experimental results verifies the theoretical model.

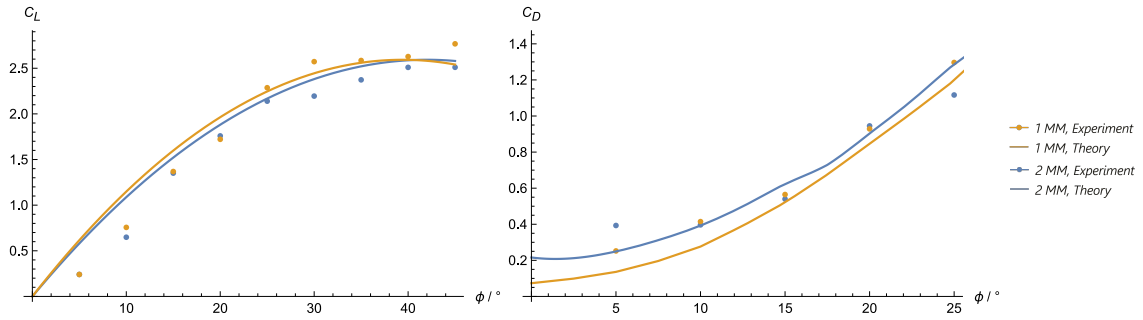


Figure 6: (left) C_L against the angle of attack ϕ . (right) C_D against the angle of attack ϕ .

When the angle of attack increases, it is observed that the lift produced also increases (Figure 6 (left)). This is due to the greater deflection of air downwards by the blades which increases the force that the air exerts on the blades and thus, increasing lift. This also increases the drag force as the projected area of the blade in the direction from which the air is flowing (Figure 6 (right)).

However, there is a decrease in the lift exerted on the blade from 40deg onwards. (Figure 6 (left)) This is because of the phenomenon of stalling. When the angle of attack exceeds a critical angle, the flow above and below the blade separates (Figure 7). This results in turbulent air flow behind the blades of the hand helicopter, resulting in greater drag and lower lift.

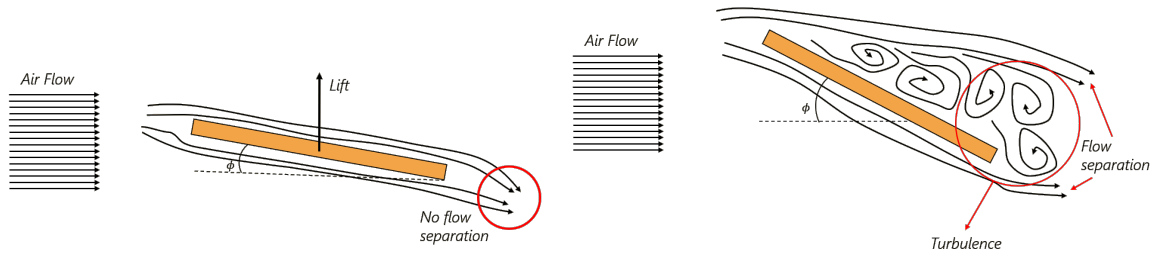


Figure 7: (left) A blade with a small angle of attack (right) A blade with a large angle of attack

The thickness of the blade also plays an important role. Our experimental results and theoretical predictions show that thinner blades experience more lift and less drag. This is also due to the effects of stalling as thinner blades experience less flow separation and thus, will produce a greater amount of lift and less drag.

3 Flight Dynamics

Having obtained the lift and drag coefficients, we will investigate the flight dynamics of the hand helicopter.

3.1 Lift and Drag

To find the lift and drag on the whole wing, we first analyse a point on the wing (Figure 8). It has a tangential velocity due to the wing's rotation, and a vertical velocity due to helicopter's movement. The net velocity is the vector sum of these 2 components. From this, we obtain the direction of the lift force,

perpendicular to the net velocity and the direction of the drag force, opposite the net velocity.

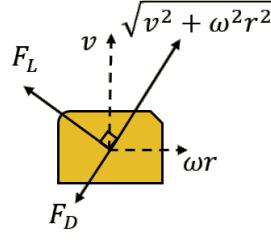


Figure 8: A diagram showing the forces acting on a cross-section of a hand helicopter's wing

Now, the lift and drag on the wing can be expressed with the following integrals.

$$F_{L,z} = \frac{1}{2} \rho C_L w \omega \int_0^l r \sqrt{v^2 + \omega^2 r^2} dr \quad (3)$$

$$F_{D,z} = \frac{1}{2} \rho C_D w v \int_0^l \sqrt{v^2 + \omega^2 r^2} dr \quad (4)$$

$$\tau_D = \frac{1}{2} \rho C_D w \omega \int_0^l r^2 \sqrt{v^2 + \omega^2 r^2} dr + \frac{1}{2} \rho C_L w v \int_0^l r \sqrt{v^2 + \omega^2 r^2} dr \quad (5)$$

Evaluating these integrals, we obtain the following analytical expressions for the lift and drag force on the hand helicopter.

$$F_{L,z} = \frac{1}{6} \rho C_L w \omega ((v^2 + \omega^2 l^2)^{\frac{3}{2}} - v^3) \quad (6)$$

$$F_{D,z} = \frac{\rho C_D w v}{4\omega} (\omega l \sqrt{v^2 + \omega^2 l^2} + v^2 \ln \frac{\omega l + \sqrt{v^2 + \omega^2 l^2}}{v}) \quad (7)$$

$$\tau_D = -\frac{\rho C_D w}{16\omega^2} (\omega l (v^2 + 2\omega^2 l^2) \sqrt{v^2 + \omega^2 l^2} - v^4 \operatorname{arcsinh} \frac{\omega l}{v}) - \rho C_L w v \frac{(v^2 + \omega^2 l^2)^{\frac{3}{2}} - v^3}{6\omega^2} \quad (8)$$

3.2 Lift-Off

These expressions allow us to obtain the minimum angular velocity required for the hand helicopter to lift-off the ground. This occurs when the lift produced by the hand helicopter is sufficient to overcome its weight. To find this angular velocity, we can equate the weight of the hand helicopter to the lift produced.

$$mg = \frac{1}{6} \rho C_L w \omega^2 l^3 \quad (9)$$

Solving this equation, an analytic expression for the minimum angular velocity for lift-off can be obtained.

$$\omega^* = \sqrt{\frac{6mg}{\rho C_L w l^3}} \quad (10)$$

3.3 Trajectory and Maximum Height

With the lift and drag forces on the hand helicopter, we can also obtain its equations of motion. Applying Newton's 2nd Law on the hand helicopter, we obtain the following coupled ordinary differential equations describing the motion of the hand helicopter.

$$m\ddot{x} = F_{L,z} - F_{D,z} - mg \quad (11)$$

$$I\ddot{\theta} = \tau_D \quad (12)$$

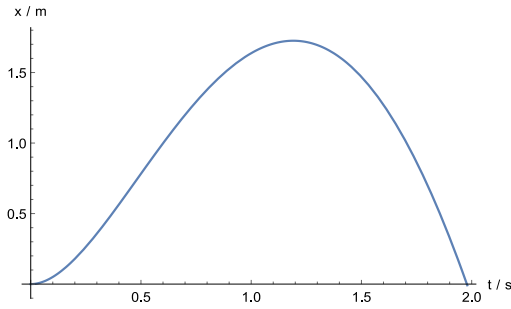


Figure 9: A graph of the hand helicopter's trajectory

After solving the equations of motion, we can obtain the following sample solution (Figure 9). Initially, the hand helicopter speeds up due to the lift generated being greater than its weight. After some time, due to the rotation of the hand helicopter slowing down due to drag, the hand helicopter generates less lift and eventually its weight becomes greater than lift, causing it to slow down. Then, the helicopter reaches its maximum height and descends.

3.4 Maximum Height

The theory assumes that the hand helicopter moves in one-dimension. As such, our experimental setup needs to be able to launch the hand helicopter consistently straight upwards. Several different designs for the launcher were considered including one similar to that built by Nakamura et al. [1]. In their study, Nakamura et al. [1] launched the hand helicopter through non-motorised means. However, this may not launch the hand helicopter straight upwards consistently. Eventually, we decided to build a design similar to that of Taylor [4].

The helicopter is placed in a special holder which is connected to a pulley by a belt. The pulley is rotated by a motor and its speed is measured by a tachometer (Figure 10 (left)). A ball bearing allows the holder to freely rotate. The holder is 3D printed and its design is adapted from Taylor's design [4]. It consists of a groove to hold the blades in place when rotating and a ramp to release the blades when it stops rotating, allowing the helicopter to lift out of the holder (Figure 10 (right)). A camera is also placed about 5m away (in order to reduce parallax error) to record the helicopter's motion and the maximum height achieved by the helicopter. Tracker is then used to track the motion of the hand helicopter. A demonstration of the experimental setup can be found in the supplementary video.



Figure 10: (left) A close-up of the experimental setup (right) A close-up of the hand helicopter placed inside a special holder

There is a good agreement between the experimental results and the theoretical predictions (Figure 11). We also observe that the maximum height increases when the initial angular velocity increases. This is because at a higher angular velocity, the hand helicopter produces more lift and it takes a longer time before the rotation of the hand helicopter slows down.

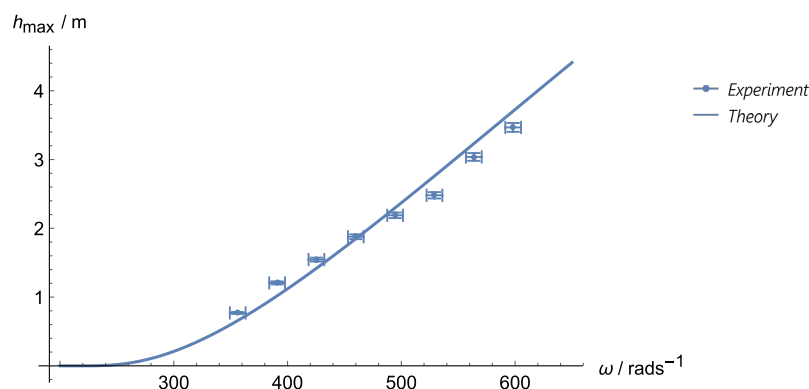


Figure 11: A graph of maximum height against the initial angular velocity

The angle of attack of the hand helicopter was varied and experiments were conducted for 20, 25 and 35 degrees (Figure 12). The maximum height increases when angle of attack increases from 20 to 25 deg due to the greater lift produced. However, the maximum height decreases when the angle of attack increases from 25 to 35 deg due to the greater amount of drag acting on the blades of the hand helicopter, causing its rotation to slow down faster. This increase in drag also causes the gradient of the graph to decrease.

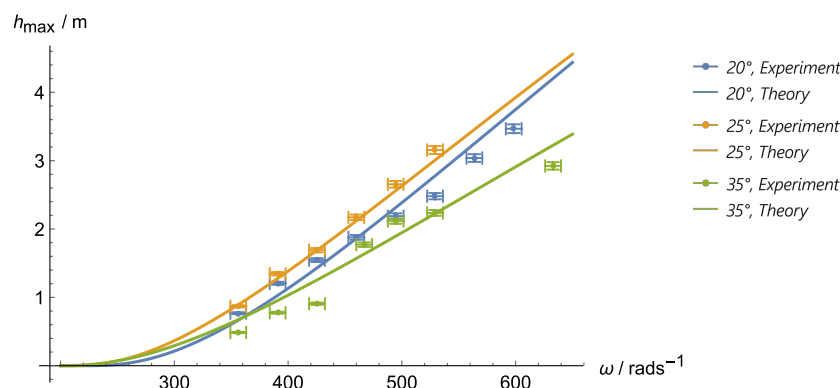


Figure 12: A graph of maximum height against the initial angular velocity for different angles of attack

Experiments were also conducted on hand helicopters with blades of 7, 8 and 9 cm lengths (Figure 13). The longer blade has a lower lift-off angular velocity (which can be seen from the x-intercept of the graph) since it generates a greater amount of lift. As the length of the blade increases, the maximum height increases due to the larger moment of inertia of the helicopter which reduces the impact of drag as well as the greater lift produced.

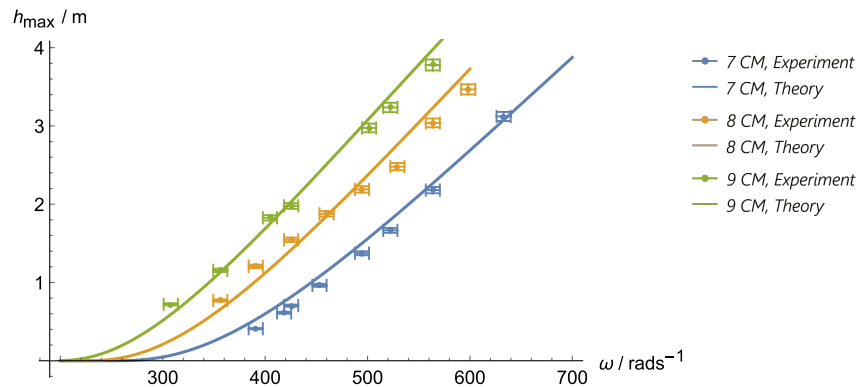


Figure 13: A graph of maximum height against the initial angular velocity of the hand helicopter for different blade lengths

Finally, the thickness of the helicopter blades was also varied and experiments were conducted on blades of thickness 1 mm and 2 mm (Figure 14). The thinner blade has a lower lift-off angular velocity as it has a lower mass and produces more lift. However, the thicker blade helicopter flies higher due to a higher moment of inertia which reduces the impact of the drag. The lower moment of inertia of the thinner blade helicopter also increases the impact of drag, making the gradient gentler.

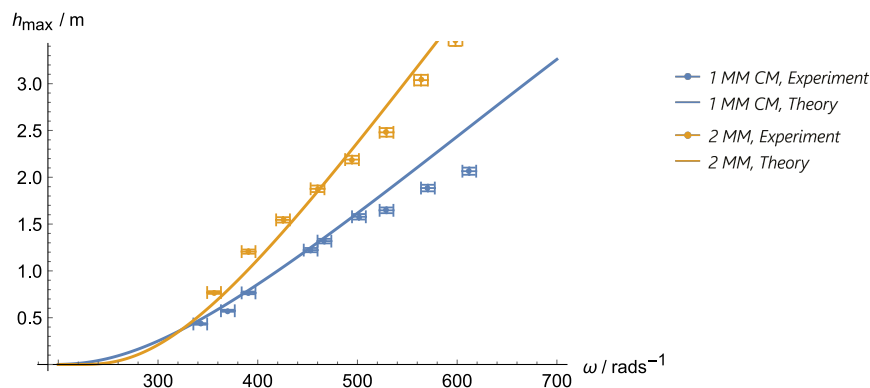


Figure 14: A graph of maximum height against the initial angular velocity for different blade thickness

3.5 Lift-Off

To investigate the minimum angular velocity required for the helicopter to lift-off, a similar setup is used (Figure 15). The helicopter is placed in a 3D printed holder which is attached to a box containing a motor. A pin is placed on the holder which pushes the helicopter to spin with the holder. When the holder is spun sufficiently quickly, the helicopter lifts up. A lubricant is applied on the pin to reduce friction. For a video of the lift-off experimental setup, please see the supplementary video.

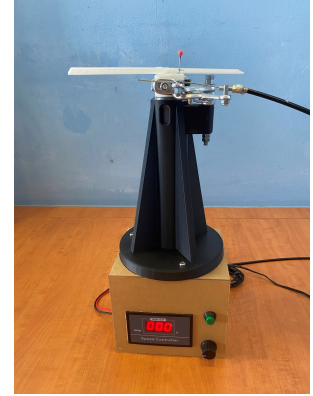


Figure 15: A picture of the lift-off angular velocity setup

There is a good agreement between the theoretically predicted minimum angular velocity for lift-off and the experimental results. There is a decrease in angular velocity as more air is deflected downwards by a blade with a larger angle of attack and more lift is produced. The rate of this decrease slows down due to the effects of stalling reducing the amount of lift produced (Figure 16). When the thickness of the blade is reduced, the lift-off angular velocity also decreases since more lift is generated by a thin blade and because the hand helicopter has a lower mass (Figure 16).

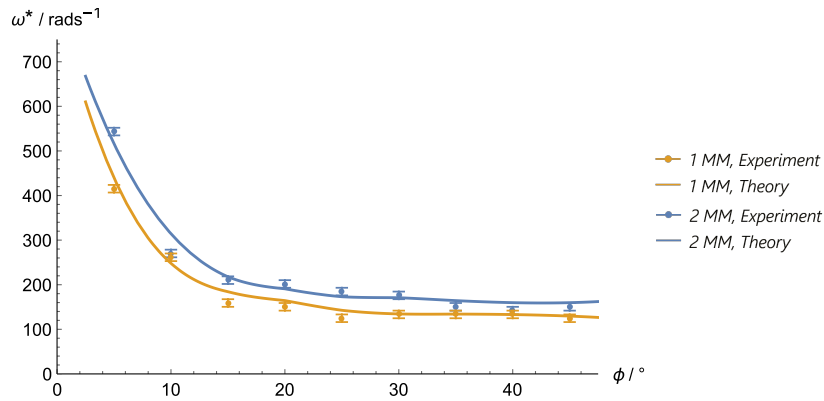


Figure 16: A graph of the minimum angular velocity for lift-off against the angle of attack

When the length of the blade increases, the lift-off angular velocity decreases as more lift is generated by the longer blades. The lift-off angular velocity tends to 0 as the length of the blade tends to infinity (Figure 17).

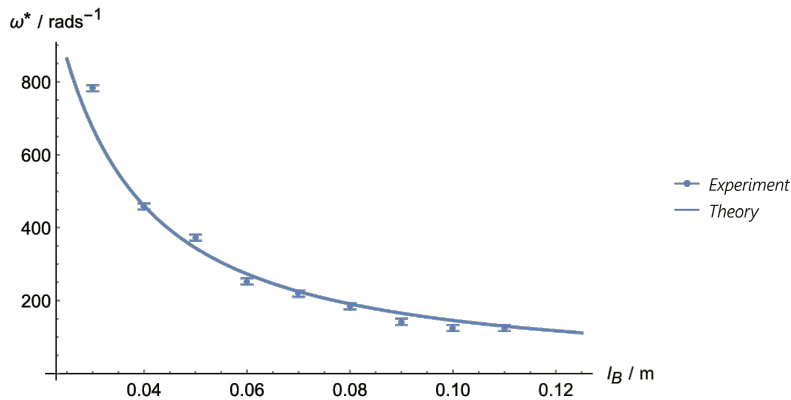


Figure 17: A graph of the minimum angular velocity for lift-off against the length of the blade

4 Stability

4.1 Tennis Racket Theorem

It is observed during experiments that when the hand helicopter's stick is too short or when the hand helicopter's blades are too long, it becomes unstable and flips over in mid air. This phenomenon can be explained with the Tennis Racket Theorem. For a video of the hand helicopter's instability, see the supplementary video.

The Tennis Racket Theorem states that rotation about the principal axis of an object with the intermediate moment of inertia is unstable while the rotation about the other 2 principal axes of the object is stable.

Looking at the moments of inertia about the hand helicopter's 3 principal axes of rotation, it becomes clear why a short stick makes the helicopter more unstable. Intuitively, as the stick gets shorter, the center of mass moves upwards. This significantly reduces the moment of inertia about the x and y axes, when the moment of inertia about the z axis remains relatively constant. As a result, once I_z exceeds I_x or I_y , the helicopter becomes unstable. Therefore, there exists a minimum stick length for the helicopter to be stable.

The conditions for stability are given by the following inequality.

$$I_x < I_z < I_y \text{ or } I_y < I_z < I_x \quad (13)$$

After finding the moments of inertia of the hand helicopter, we can predict the stability of the hand helicopter.

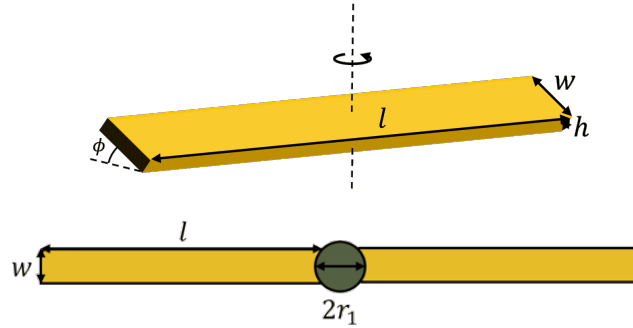


Figure 18: (left) A diagram of a single hand helicopter blade (right) A diagram of the hand helicopter blades / propeller

The moment of inertia of a single blade (about its centre of mass) and the blades (Figure 18) has the following expression.

$$I_B = \frac{m_B}{12}(w^2 \cos^2 \theta + h^2 \sin^2 \theta + l^2) \quad (14)$$

$$I_P = n \left(I_B + m_B \left(\frac{l}{2} + r_1 \right)^2 \right) + \frac{1}{2} m_c r_1^2 \quad (15)$$

Adding in the moment of inertia of the stick, we have

$$I_z = \frac{1}{2} m_s (r_{int}^2 + r_{ext}^2) + I_P \quad (16)$$

The principal axes of the hand helicopter intersect at this centre of mass (Figure 19), whose position can be calculated as follows

$$x_{CM} = \frac{m_P l_P + m_s \left(l_P + \frac{l_s}{2} \right)}{nm_B + m_C + m_s} \quad (17)$$

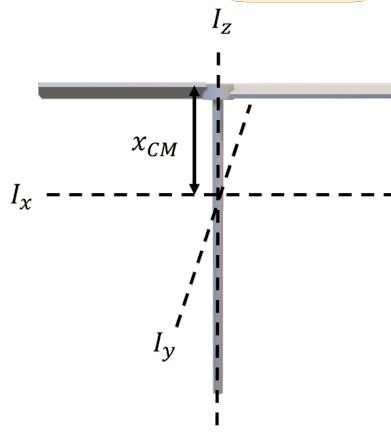


Figure 19: A diagram of the hand helicopter and its 3 principal axes

Now, we find the moments of inertia about the x-axis and y-axis. Applying parallel axis theorem and using well-known moments of inertia formulas, we obtain these expressions.

$$I_x = \frac{1}{4}m_s (r_{int}^2 + r_{ext}^2) + \frac{1}{12}m_s l_s^2 + m_s \left(l_P^2 + \frac{l_s}{2} - x_{CM} \right)^2 + \frac{1}{4}m_c r_1^2 + \frac{1}{12}m_c l_P^2 + m_c \left(\frac{l_P}{2} - x_{CM} \right)^2 + \frac{n}{6}m_B (w^2 + h^2) + nm_B \left(\frac{l_P}{2} - x_{CM} \right)^2 \quad (18)$$

$$I_y = \frac{1}{4}m_s (r_{int}^2 + r_{ext}^2) + \frac{1}{12}m_s l_s^2 + m_s \left(l_P^2 + \frac{l_s}{2} - x_{CM} \right)^2 + \frac{1}{4}m_c r_1^2 + \frac{1}{12}m_c l_P^2 + m_c \left(\frac{l_P}{2} - x_{CM} \right)^2 + \frac{n}{6}m_B (l^2 + h^2) + nm_B (l^2 + x_{CM}^2) \quad (19)$$

Thus, with the expressions for the moment of inertia of the hand helicopter, we can find the conditions for the hand helicopter to be stable.

4.2 Experimental Comparison

To investigate the stability of the helicopter, we 3D printed various lengths of blades (3 - 12 cm) and sticks (5 - 25 cm) of the hand helicopter.

To determine if the hand helicopter is stable, we launch it by hand and see if it is able to fly properly or if it flips over in mid-air.

We then plot a phase plot of the length of the hand helicopter's blade against the length of the hand helicopter's stick. The green (stable) and red (unstable) regions are obtained from solving the inequality (equation 13). The dots in the phase plot represent the experimental data. There is a good agreement between the theoretical predictions and the experimental results. This shows that the cause of the instability of the hand helicopter at shorter stick lengths and longer blade lengths can be explained by the Tennis Racket Theorem.

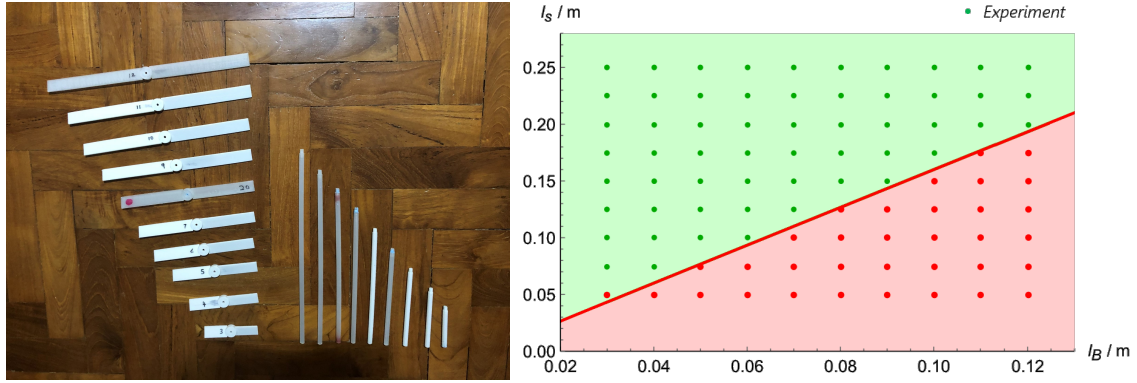


Figure 20: (left) A picture of various lengths of 3D printed blades and sticks (right) A phase plot of the hand helicopter's stability (length of blade against stick length).

5 Conclusion and Future Work

In conclusion, we have predicted the lift and drag coefficients of the hand helicopter using a 3 dimensional computational fluid dynamics simulation and obtained a good agreement with our experimental data. The impact of the thickness of the blade and the angle of attack of the blade on the lift and drag coefficients have also been analysed. The analysis has shown that increasing the angle of attack, increases lift and drag until a certain critical angle, at which flow separation occurs, causing the lift to decrease. In future, the effects of the width of the helicopter blades and the blades geometry would also be interesting to investigate. For instance, using an blade with an elliptical cross-section could increase the lift and decrease the drag produced.

The lift-off angular velocity and maximum height of hand helicopter have also been theoretically predicted and experimentally verified. The experimental results have shown that there are many factors affecting the maximum height such as the lift and drag acting on the hand helicopter as well as the hand helicopter's mass and moment of inertia. This makes optimising the maximum height a very interesting problem. For instance, while decreasing the length of the stick will reduce the hand helicopter's mass and thus allow it to fly higher, it will also reduce the hand helicopter's stability. When changing the angle of attack, there also exists an optimum angle of attack which balances increase in maximum height provided by lift and the decrease in maximum height as a result of increased drag. There also exists an optimum thickness of the blade which balances the increase in lift, reduction in drag and mass with the reduction in the moment of inertia from reducing the blade's thickness. As such, the optimisation of maximum height of the hand helicopter is an area which merits future investigation.

Finally, the stability of the hand helicopter was analysed and the cause of the instability was determined to be the Tennis Racket Theorem. The theoretical predictions were compared with the experimental results, obtaining a good agreement. However, there is still more to be investigated. Other parameters

regarding the stability of the hand helicopter can be analysed such as the maximum height of the flight of the unstable helicopter and the rate of precession of a stable hand helicopter. This could possibly be done by finding the equations describing the full 3D motion of the hand helicopter.

Declaration of Past Submission

This project was previously submitted for the following science fairs / competitions:

Competition	Date	Awards Won
Singapore Science and Engineering Fair 2022	Jan - Mar 2022	Bronze
National STEM Talent Search 2022	Jul 2022	Participation

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6 Appendix

6.1 Supplementary Materials

[Supplementary Video 1](#) - Experimental Setup and Hand Helicopter's Flight and Instability

6.2 Mesh Size Statistics

Statistic	Value
Mesh Vertices	3226
Tetrahedra	16843
Pyramids	420
Prisms	336
Triangles	900
Edge Elements	226
Vertex Elements	52

Table 1: Statistics of the mesh used for the CFD simulation